Mathematics A

Unit 2 • Chapter OL-T15 Classified

Cross-session • chapter-organised candidate

11 classified items • 37 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 2 • Unit 2 • Chapter OL-T15 • 11 items • 37 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Direct & Inverse Proportion

Unit 2 • 37 marks • 11 item(s)

June 2025 | 4MA1-1H Q18 | 4 marks

Recover the input under inverse variation

June 2024 | 4MA1-2H Q17 | 3 marks

Root direct variation y=ksqrtx

June 2023 | 4MA1-2H Q16 | 5 marks

Root direct variation y=ksqrtx

November 2024 | 4MA1-1H Q14 | 3 marks

Inverse-square or inverse-power variation

June 2022 | 4MA1-2H Q17 | 4 marks

Power direct variation y=kx^n

January 2022 | 4MA1-1H Q15 | 3 marks

Recover the input under inverse variation

June 2021 | 4MA1-2H Q16 | 3 marks

Inverse-square or inverse-power variation

June 2021 | 4MA1-2H Q16 | 3 marks

Inverse-square or inverse-power variation

May-Nov 2020 | 1H Q14 | 3 marks

Inverse-square or inverse-power variation

January 2023 | 1H Q17 | 3 marks

Inverse-square or inverse-power variation

November 2025 | 2H Q17 | 3 marks

Inverse-square or inverse-power variation

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

18 y is inversely proportional to x^3

$y = 4$ when $x = 3$

Work out the value of x when $y = 864$

$x =$ ....

(Total for Question 18 is 4 marks)

WORKING SPACE

Question 22

4MA1-2H Q17

MODULAR UNIT 2

3 marks

17 Q is directly proportional to the square root of d

$Q = 4.5$ when $d = 324$

Find a formula for Q in terms of d

....

WORKING SPACE

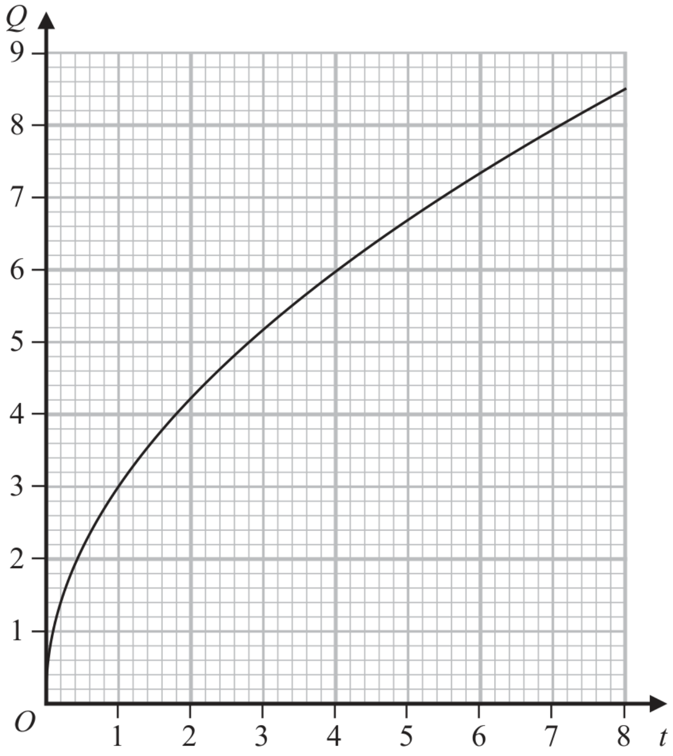
Question 26

4MA1-2H Q16

MODULAR UNIT 2
5 marks

16 Q is directly proportional to $\sqrt{t}$

The graph shows the relationship between Q and t for $0 < t < 8$



(a) Find a formula for Q in terms of t

....
(3)

Q is increased by 20%

(b) Find the percentage increase in t

....%
(2)

Working space for Question 26

Continue your method

UNIT 2
5 marks

WORKING SPACE

Question 11

4MA1-1H Q14 M01

MODULAR UNIT 2

3 marks

14 B is inversely proportional to the square of d

$B = 0.25$ when $d = 12$

Find a formula for B in terms of d

....

WORKING SPACE

Question 29

4MA1-2H Q17 Q17

MODULAR UNIT 2
4 marks

17 M varies directly as the cube of h

$M = 4$ when $h = 0.5$

Find the value of h when $M = 500$

....

WORKING SPACE

Question 10

4MA1-1H Q15

MODULAR UNIT 2

3 marks

15 A is inversely proportional to C^2

$A = 40$ when $C = 1.5$

Calculate the value of C when $A = 1000$

$C =$ ....

WORKING SPACE

16 A is inversely proportional to the square of r

$A = 5$ when $r = 0.3$

(a) Find a formula for A in terms of r

....

(3)

(b) Find the value of A when $r = 7.5A$

$A =$ ....

(3)

YOUR WORKING

16 A is inversely proportional to the square of r

$A = 5$ when $r = 0.3$

(a) Find a formula for A in terms of r

....
(3)

(b) Find the value of A when $r = 7.5A$

$A =$ ....
(3)

YOUR WORKING

May-Nov 2020 | 1H | Question 14 | 3 marks

Chapter OL-T15: Direct & Inverse Proportion

14 F is inversely proportional to the square of v.

Given that $F = 6.5$ when $v = 4$

find a formula for F in terms of v.

....

(Total for Question 14 is 3 marks)

YOUR WORKING

January 2023 | 1H | Question 17 | 3 marks

Chapter OL-T15: Direct & Inverse Proportion

17 y is inversely proportional to $\sqrt{x}$

$y = c^4$ when $x = c^2$ where c is a positive constant.

Find a formula for y in terms of x and c
Give your answer in its simplest form.

....

(Total for Question 17 is 3 marks)

YOUR WORKING

November 2025 | 2H | Question 17 | 3 marks

Chapter OL-T15: Direct & Inverse Proportion

17 T is inversely proportional to $\sqrt{m}$

$T = 15$ when $m = 36$

Find a formula for T in terms of m

....

(Total for Question 17 is 3 marks)

YOUR WORKING